

VISION Resistivity
Dual Frequency
Recorded Mode

Schlumberger

Recorded Mode

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

Longitude: 3° 26' 36.13" E

Block: 31/5

UWID:

Rig Name:

Rig Type:

West Hercules

Semi-Submersible

FL: Norwegian North Sea

FL1: Section: 26 in.

FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level

Ground Level: 307.00 m

Acquisition Dates: 05-Dec-2019 -- 07-Dec-2019

Log Interval: 341.00(m) -- 932.00(m)

Index Types: Measured Depth

Index Scales: 1:200

Depth Source: Driller's Depth

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019

Other Services:

Direction & Inclination (Surveys)

Directional Drilling

Annular Pressure While Drilling

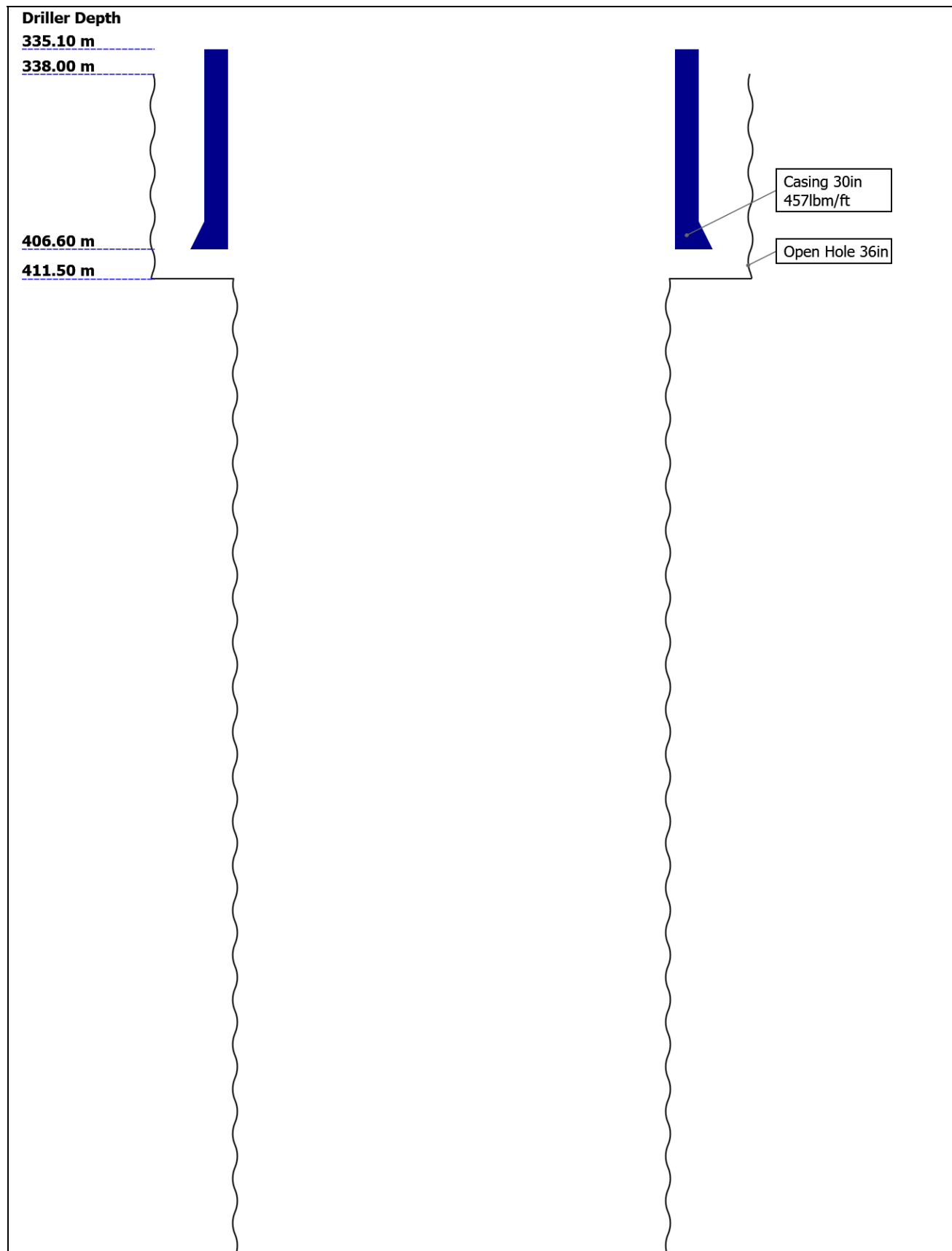
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Well Sketch





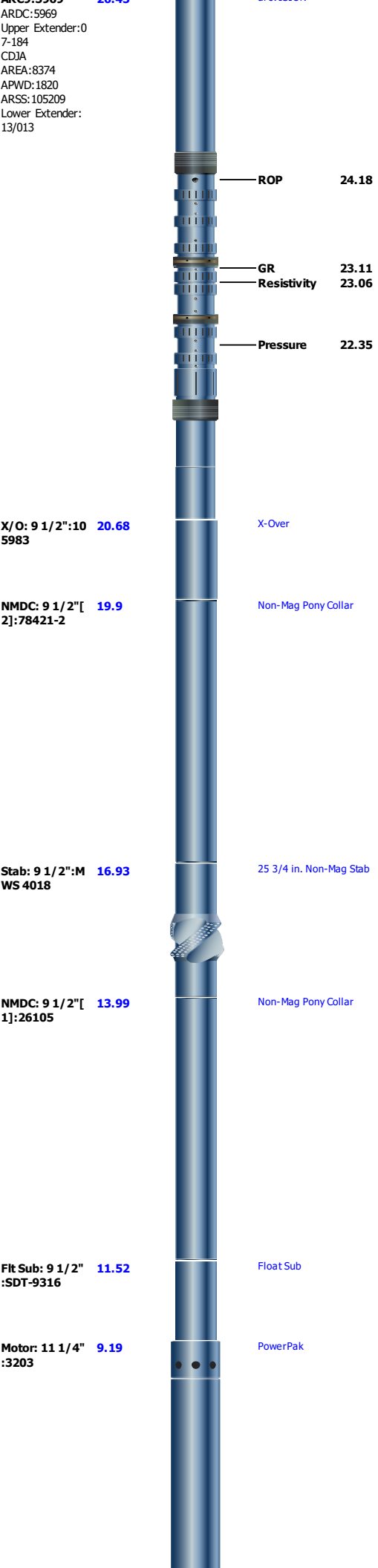
Borehole Size/Casing Record						
Bit						
Bit Size (in)	36	26				
Top Driller (m)	338	411.5				
Bottom Driller (m)	411.5	931				
Casing						
Size (in)	30					
Weight (lbm/ft)	457					
Inner Diameter (in)	27.067					
Grade	X56					
Top Driller (m)	335.1					
Bottom Driller (m)	406.6					

Operational Run Summary						
Parameter (unit)	Run 3					
Date Log Started	05-Dec-2019					
Time Log Started	01:03:12					
Date Log Finished	07-Dec-2019					
Time Log Finished	02:10:48					
Bit Size (in)	26.000					
Bit Start Depth (m)	411.50					
Bit Stop Depth (m)	931.00					
Top Log Interval (m)	406.60					
Bottom Log Interval (m)	906.80					
Max Hole Deviation (deg)	1.12					
Azimuth of Max Deviation (deg)	120.28					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	J.Johansen					
Witnessed By	S.Vikra					
Service Order Number	19SCA0085					

Borehole Fluids

Parameter(unit)	Run 3					
Fluid Type	Water					
Fluid Name	Seawater					
Max Recorded Temperatures (degC)	16					
Source of Sample	Active Tank					
Salinity (ppm)	33579.09					
Density (g/cm3)	1.03					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.19 @ 23.9					
RMF @ Meas Temp (ohm.m@degC)	0.19 @ 23.9					
RMC @ Meas Temp (ohm.m@degC)						
RM @ BHT (ohm.m@degC)	0.23 @ 16					
RMF @ BHT (ohm.m@degC)	0.23 @ 16					
RMC @ BHT (ohm.m@degC)	NaN @ 16					
Total Solid (%)						
High Gravity Solids (%)						

Run 3: Toolstring				Run 3: Remarks
Equip name TELE900:G5093	Length 35.13	MP name TeleScope	Offset	All depths referenced to driller's depth. See EOWR for depth tracking details.
MSSU900:71539 Upper Extender:2 6-02-16 MDC900:G5093 MMA:2068 MDI:3956 PMGR:455 PMEA:3708 MTA:2526 MTK900:008/298 895-2142 MSSD900:973247 -1 Lower Extender: 12/142				All data from tool memory. All data acquired while drilling.
				Gamma Ray measurement is corrected for collar thickness only.
				Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
				Phase Resistivity 16 and 22 in. spacing adversely affected by large borehole corrections.
				Run 3 and 26 in. section TD at 931 m.
		D&I	30.56	
		GR	29.91	
		ROP	28.21	
ARC9:5969	26.45	arcVISION		





Bit: 26":RK314 0.74
8

Smith Insert GS04B TCI

TOOL_ZERO

Lengths are in m

Maximum Outer Diameter = 26.000 in

Line: Sensor Location, Value: Gating Offset

All measurements are relative to TOOL_ZERO

Run 3

VISION Resistivity - Dual Frequency

Software Version

Acquisition System			Version	
Maxwell 2019.2			9.2.113335.3100	
Application Patch			DnM_NPD-KAI-2019.2_9.2.114235	
Computation	Description			Version
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family			9.2.113335.3100
ARC9GammaRayComputat ion	ARC9 Gamma Ray Computation Package for both Real-time and Recorded Mode			9.2.113335.3100
Tool Interface	System Version		Loaded Version	
HSPM	2019.0c.001		9.2.113335.3100	
Tool Elements	Description		Software Version	Firmware Version
DRILLING_SURFACE	DRILLING_SURFACE		9.2.113335.3100	
ARDC	ARC 9.00 Inch Tool Drilling Collar		9.2.113335.3100	V9.7B

Pass Summary

Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 3	Drilling	Down	33.15 m	930.00 m	05-Dec-2019 1:03:12 AM	07-Dec-2019 2:10:48 AM	Yes
All depths are referenced to toolstring zero							

Log

Company:Equinor Well:31/5-7

Run 3: Drilling:S010

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:200 Index Unit: m Index Type: Measured Depth Creation Date: 12-Dec-2019 15:12:05

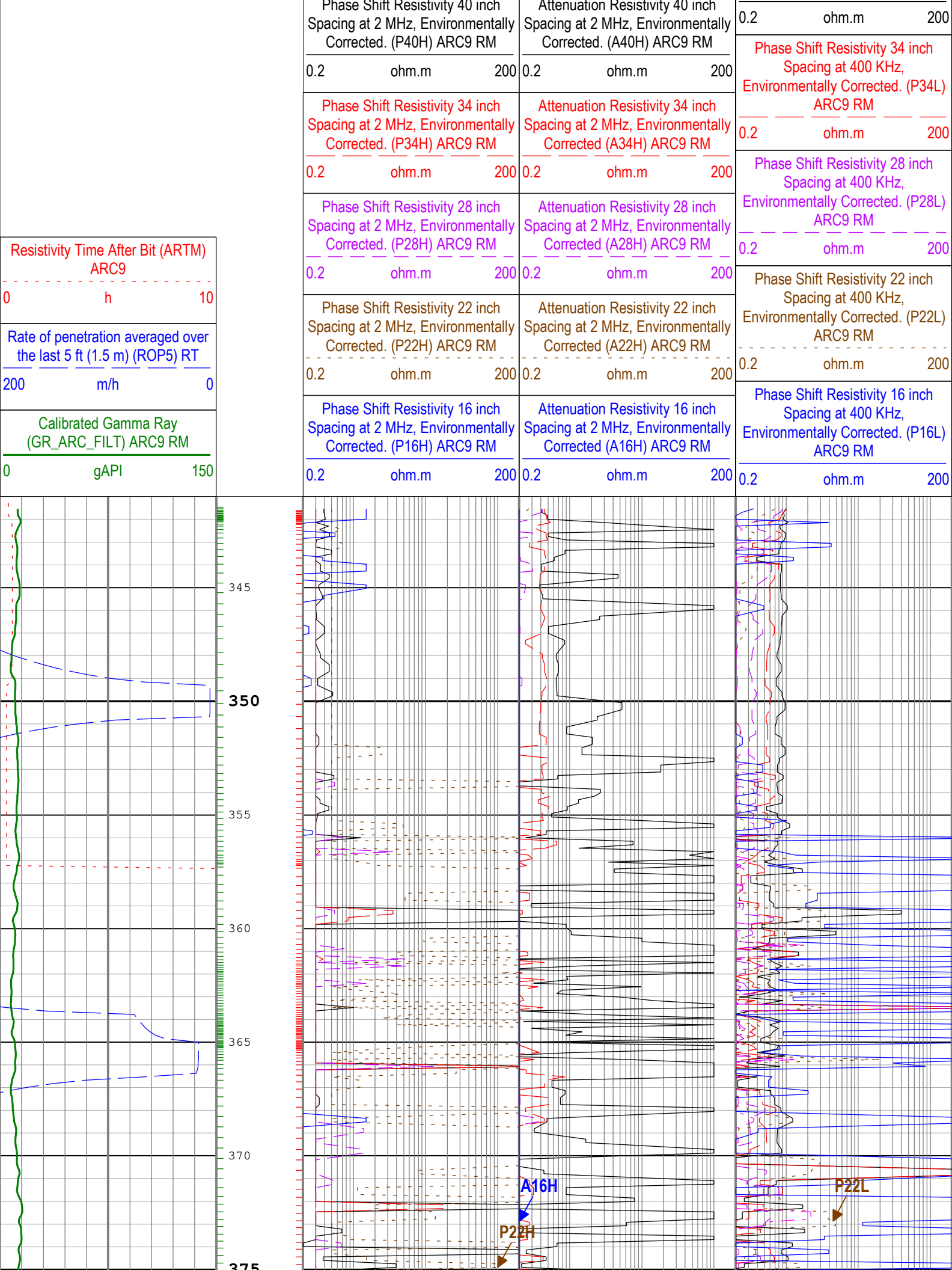
└─TICK_ARC_RES - Resistivity Tick Marks ARC9 RM

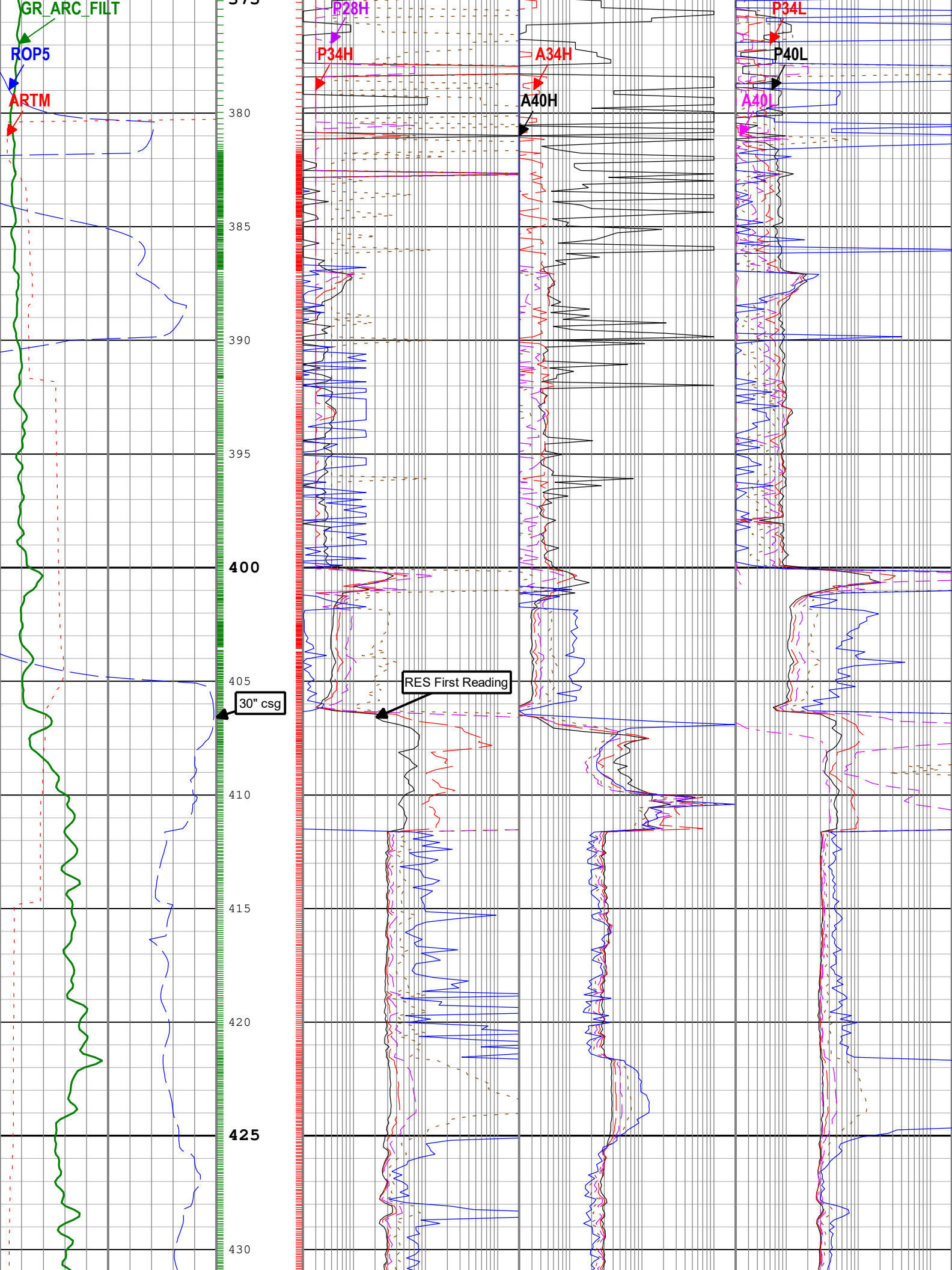
└─TICK_ARC_GR - Gamma Ray Tick Marks ARC9 RM

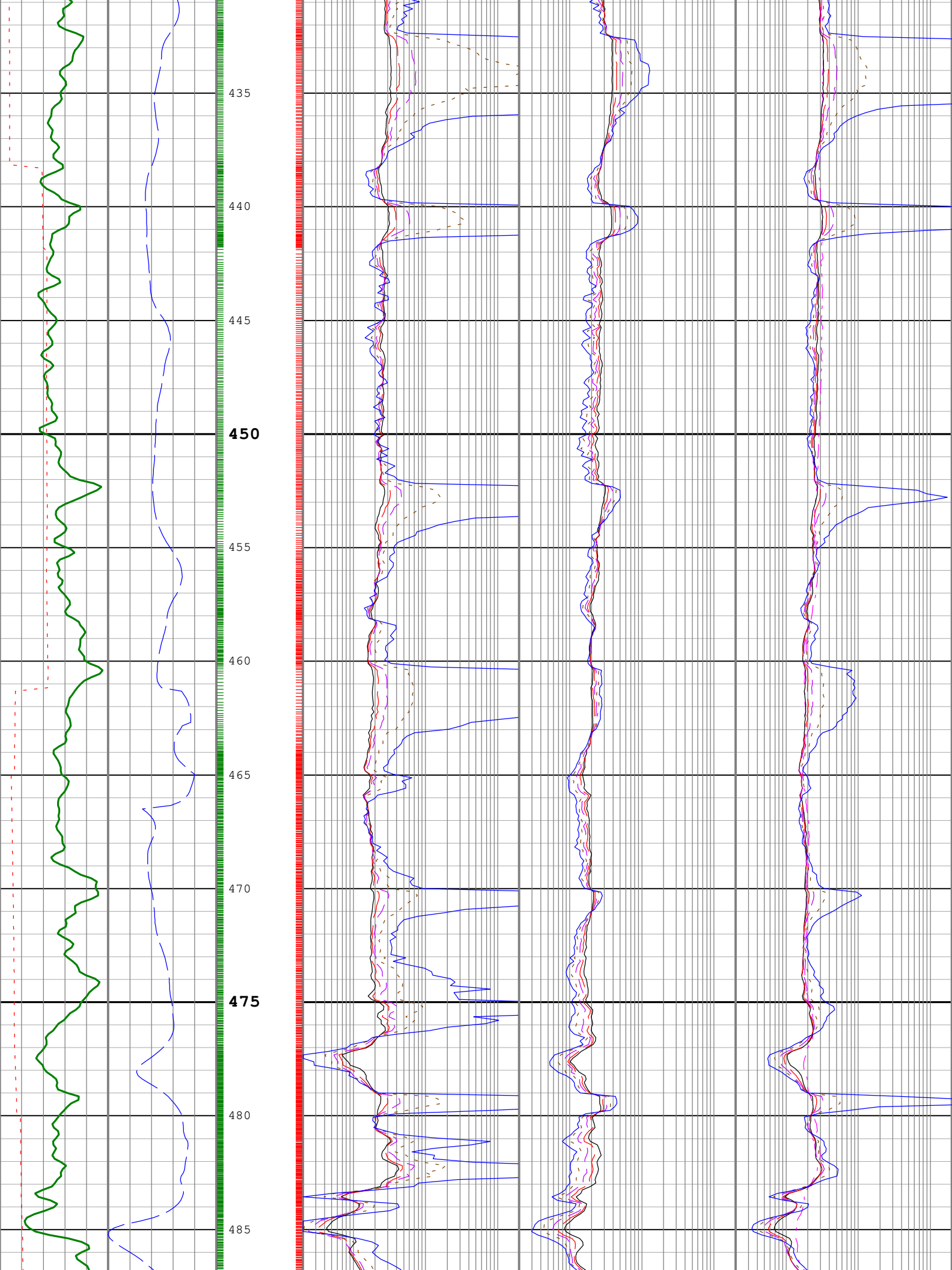
Attenuation Resistivity 40 inch
Spacing at 400 KHz,
Environmentally Corrected (A40L)
ARC9 RM

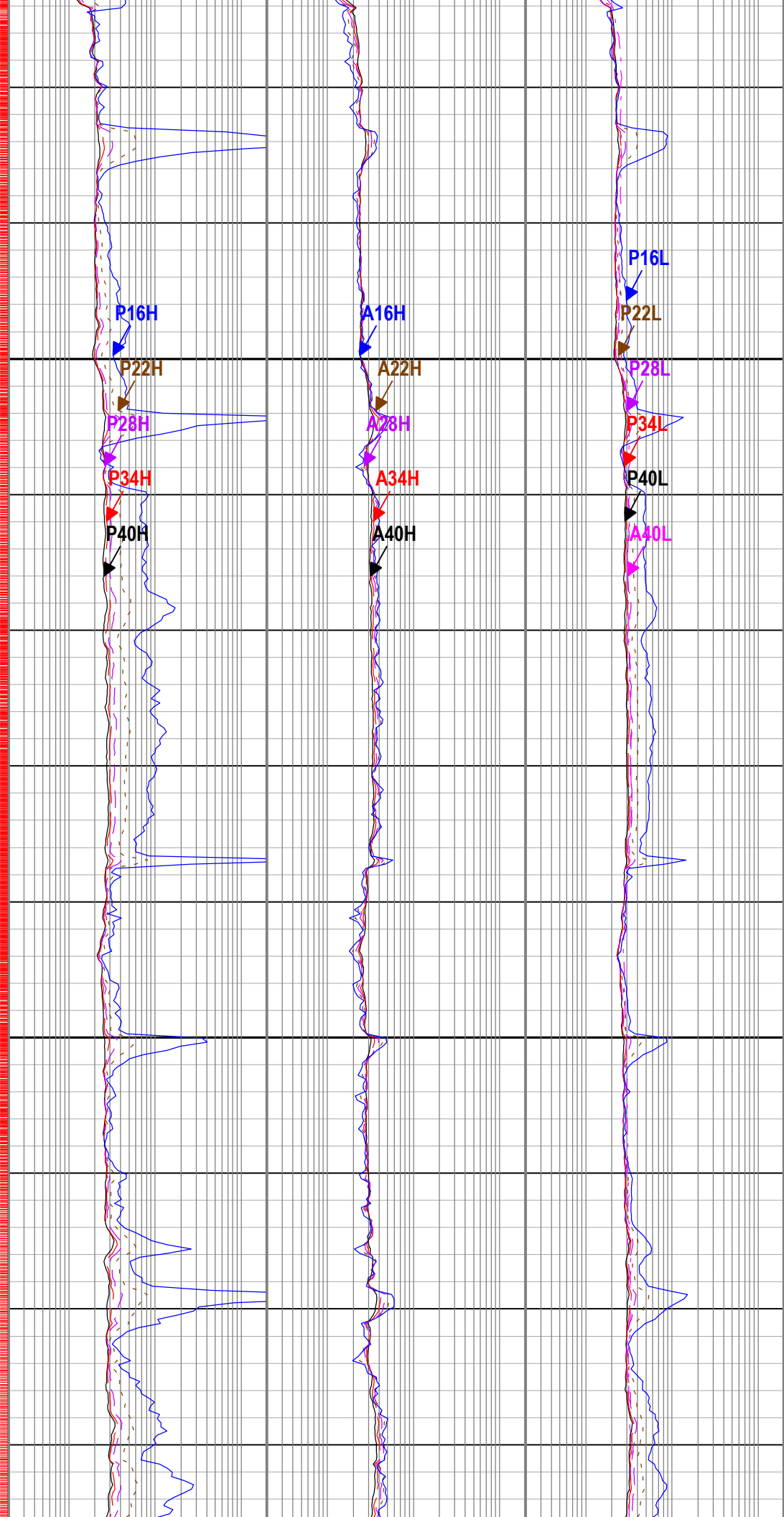
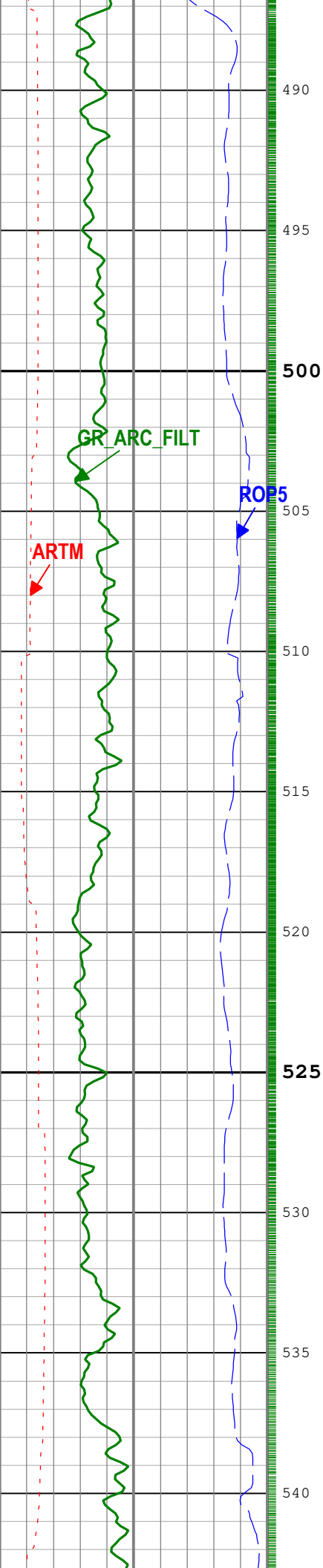
0.2 ohm.m 200

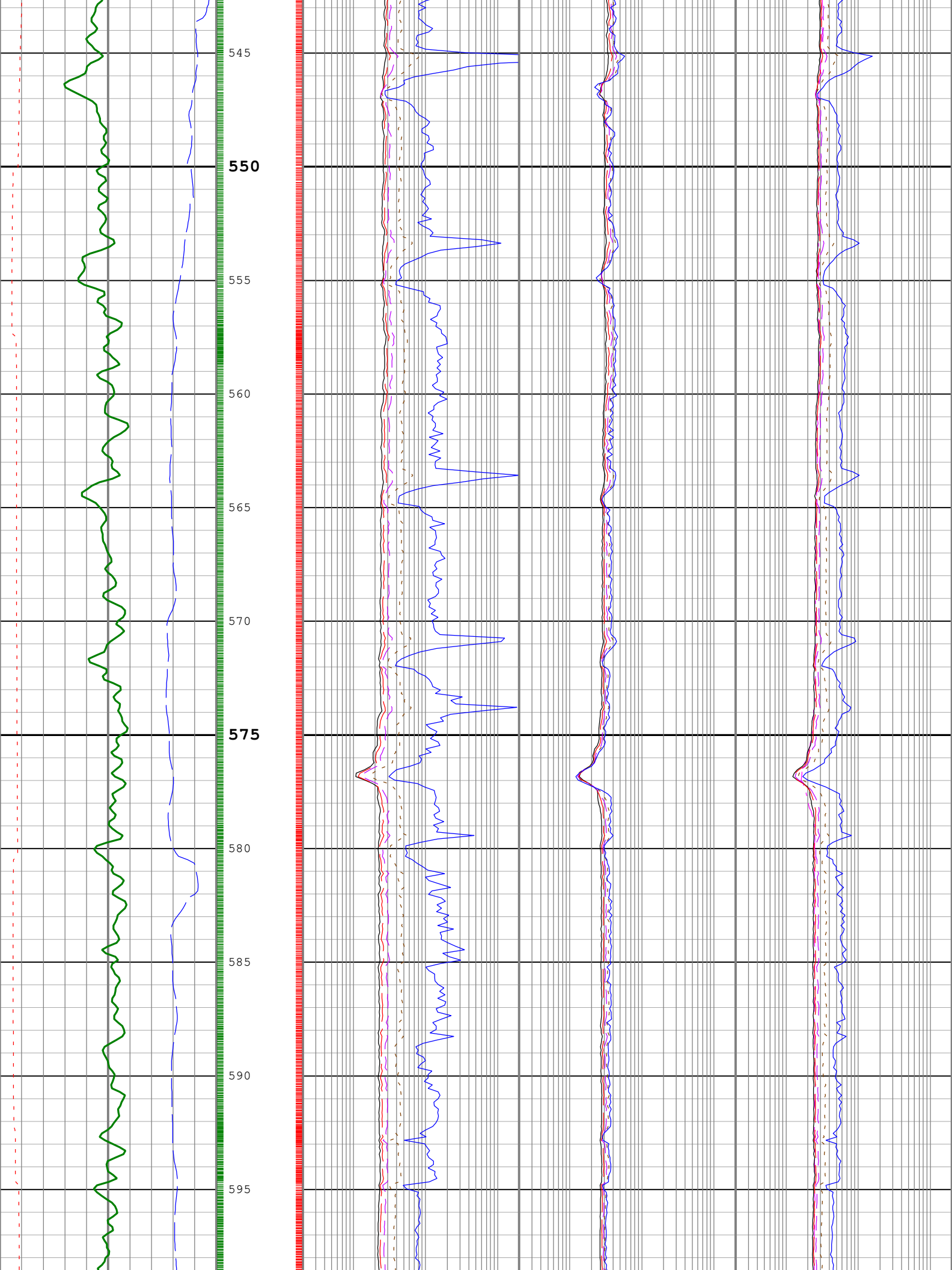
Phase Shift Resistivity 40 inch
Spacing at 400 KHz,
Environmentally Corrected. (P40L)
ARC9 RM

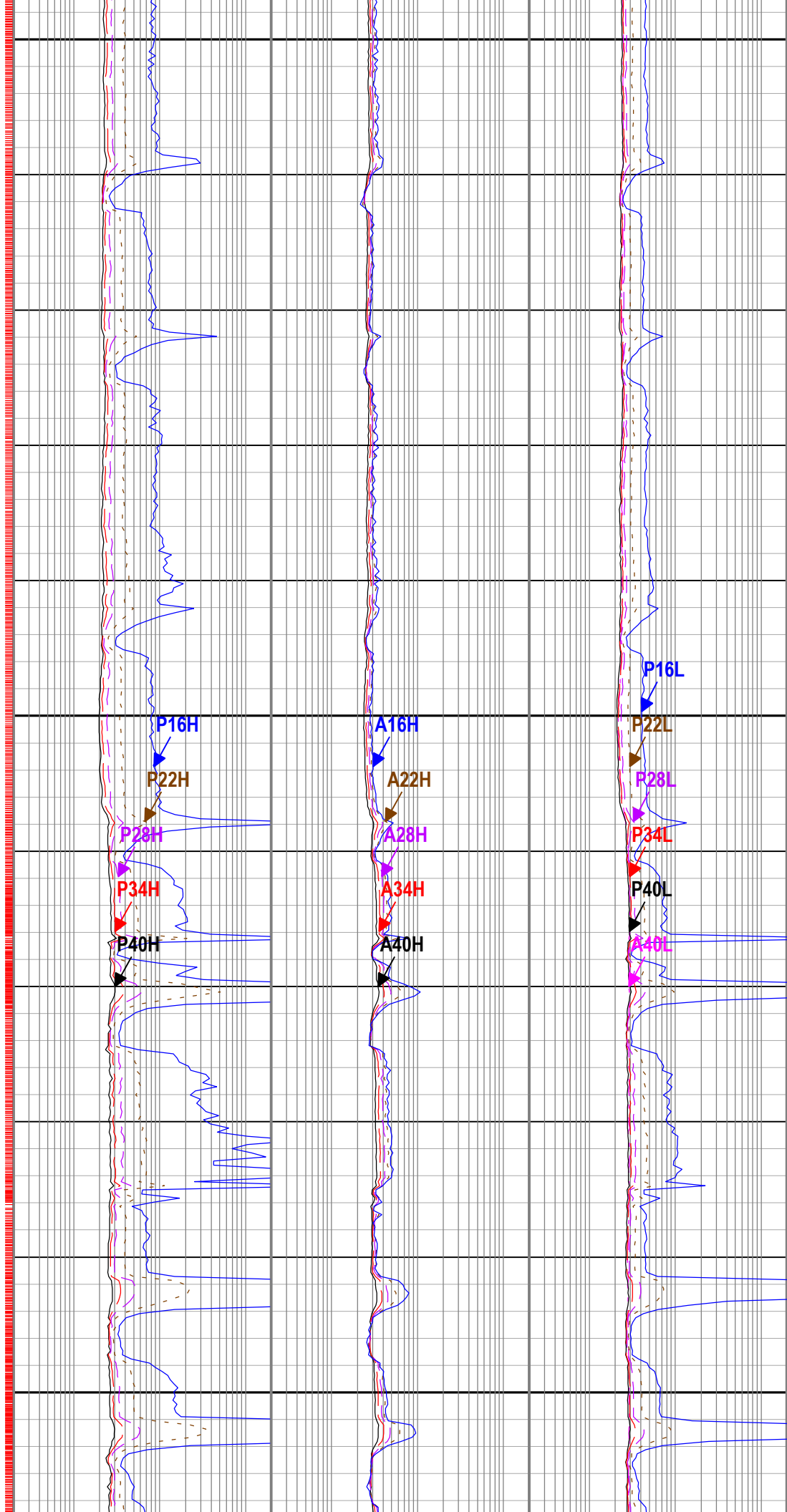
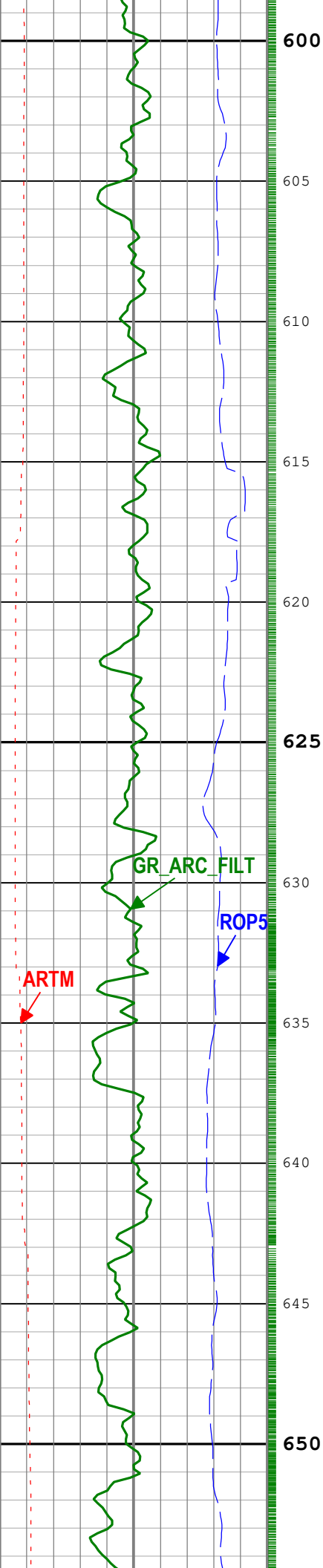


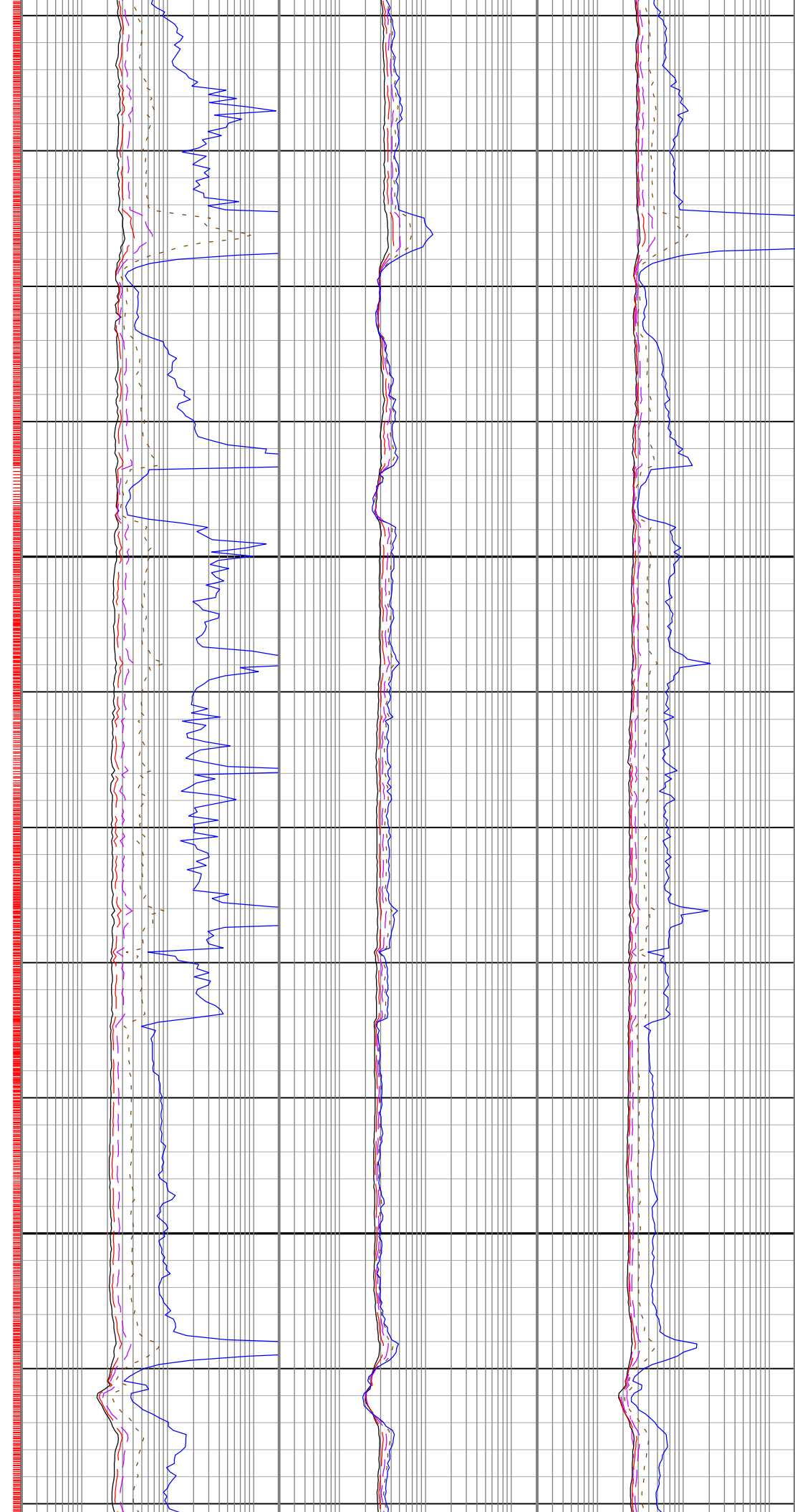
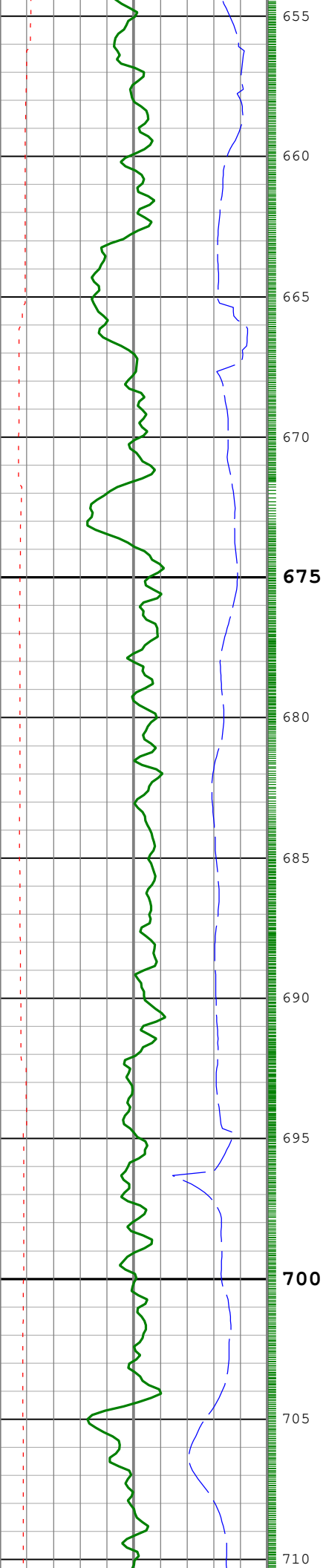


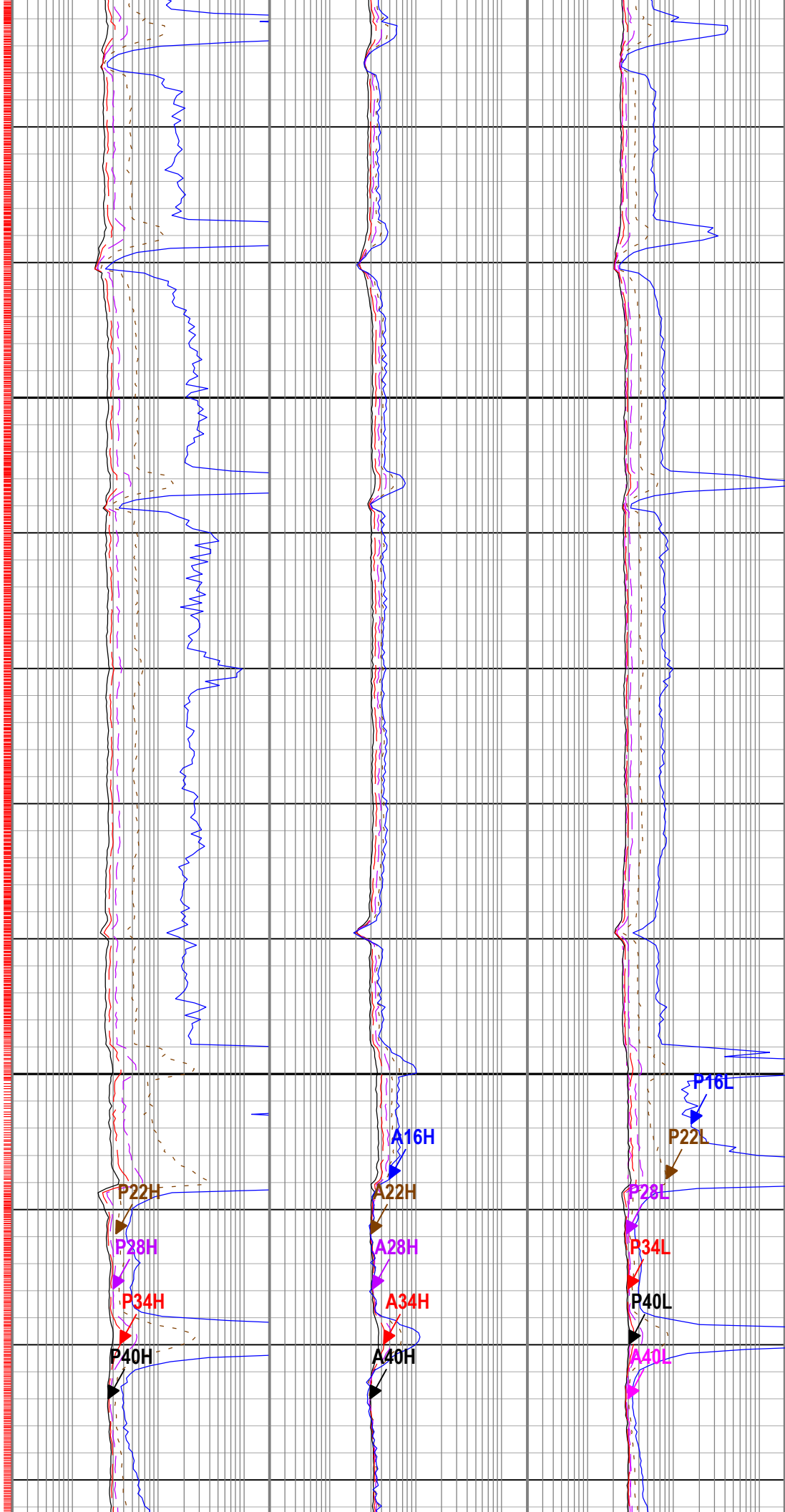
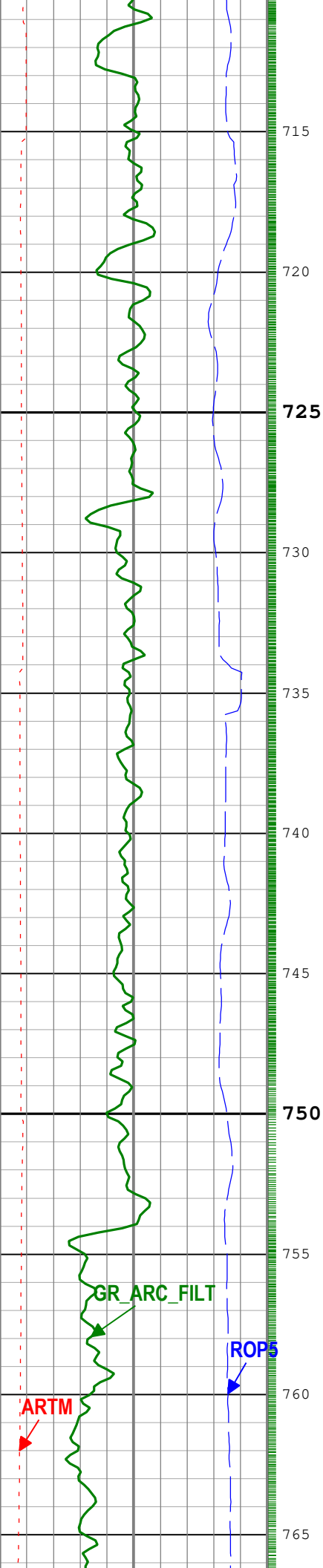


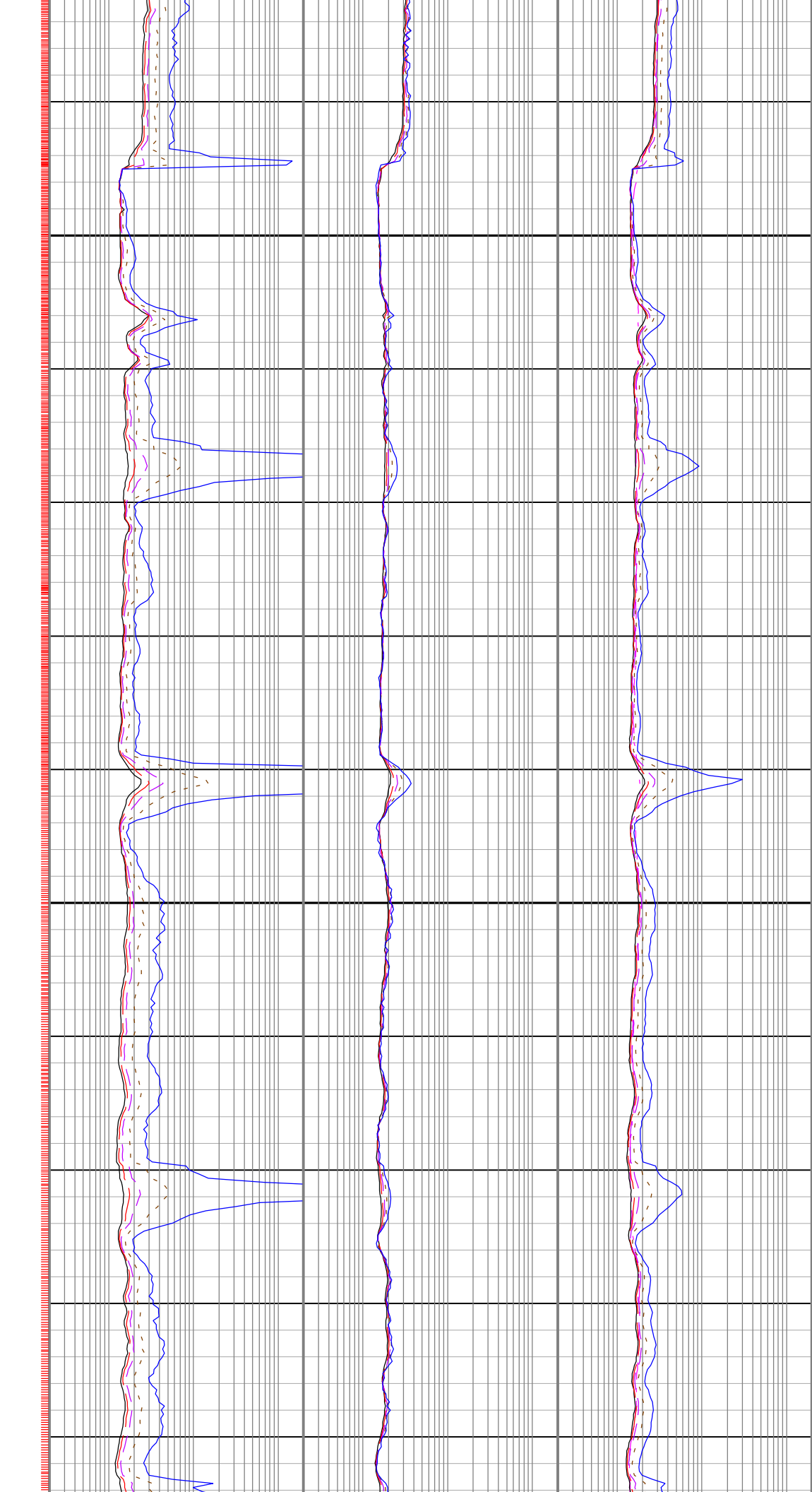
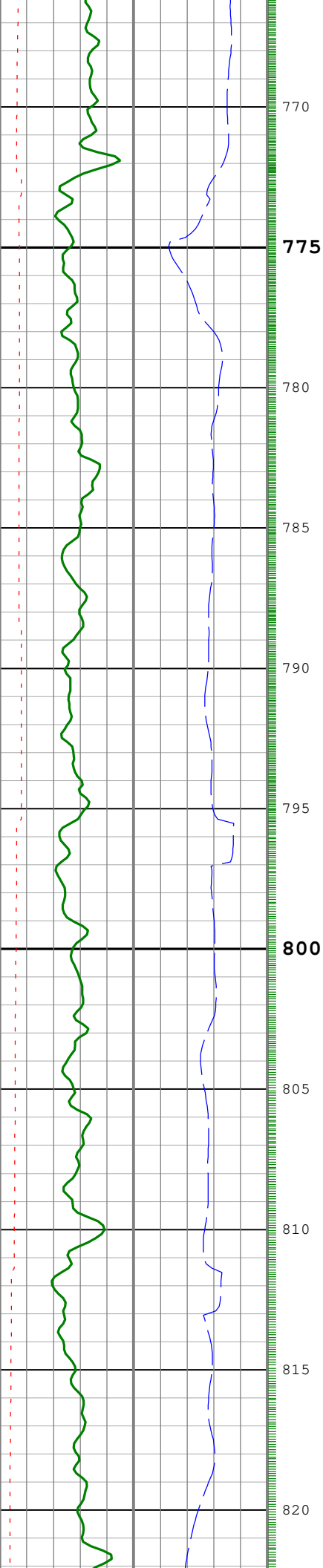


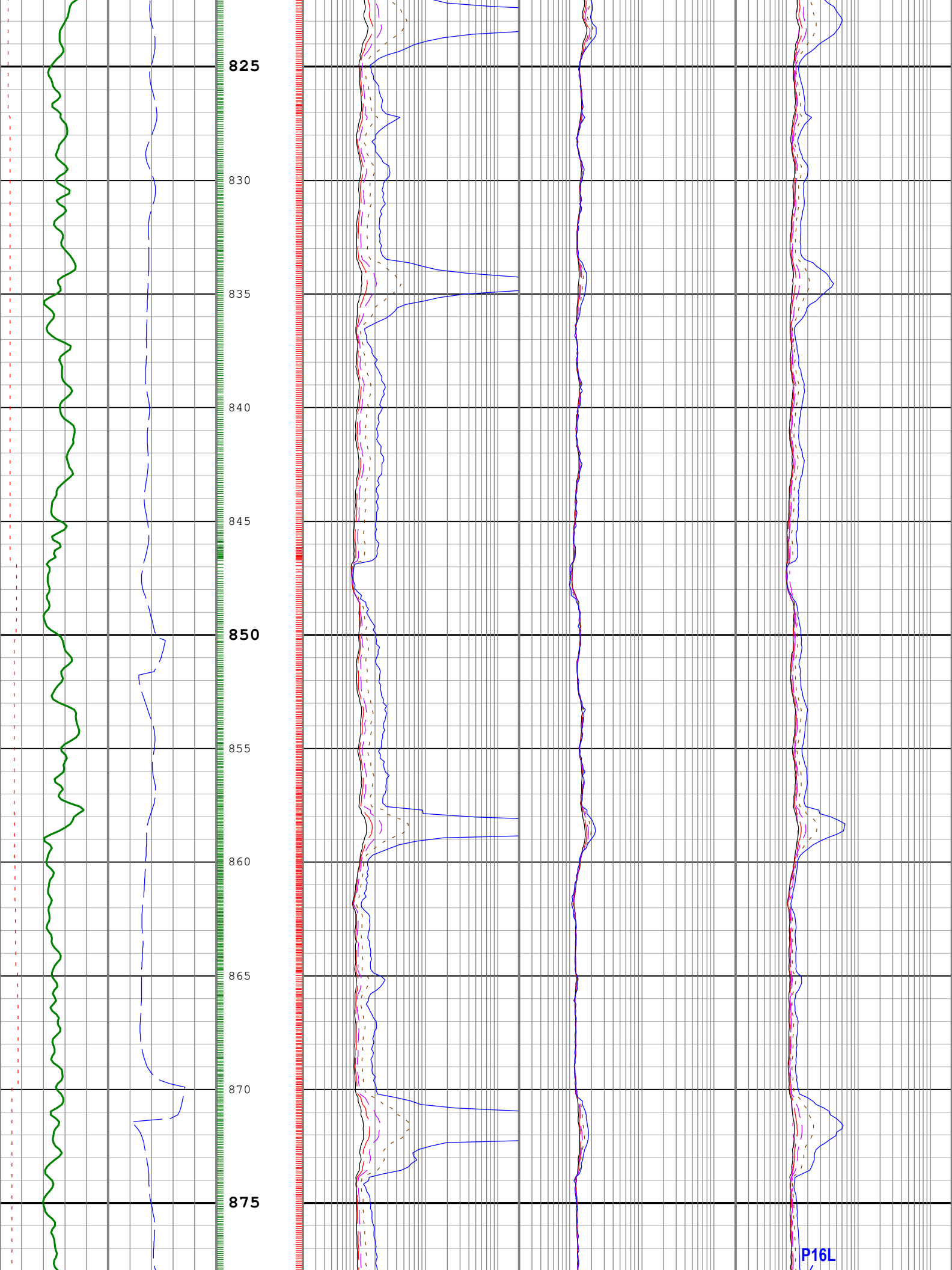


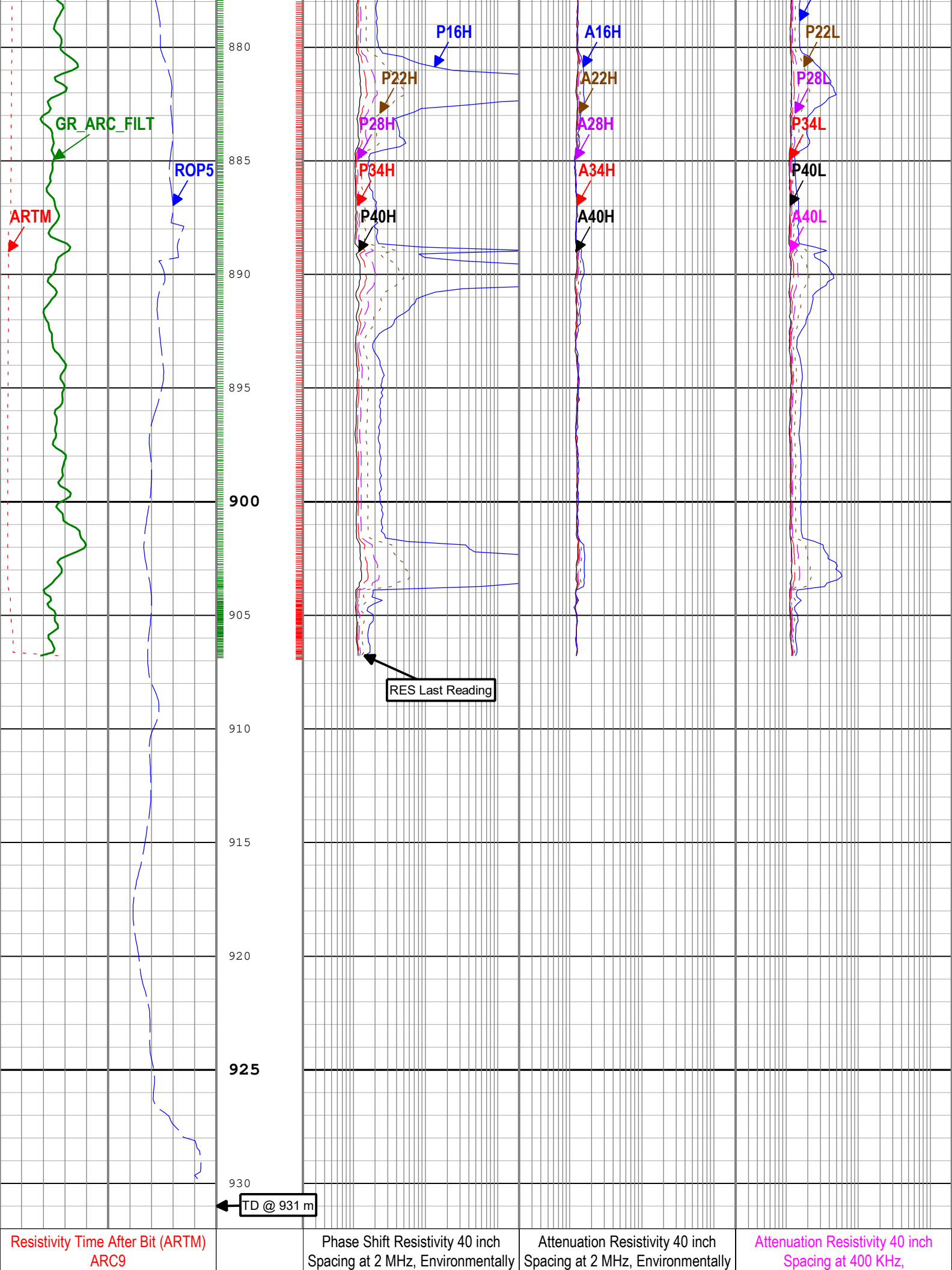












0	h	10
Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT		
200	m/h	0
Calibrated Gamma Ray (GR_ARC_FILT) ARC9 RM		
0	gAPI	150

Corrected. (P40H) ARC9 RM			Corrected. (A40H) ARC9 RM		
0.2	ohm.m	200	0.2	ohm.m	200
Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM			Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM		
0.2	ohm.m	200	0.2	ohm.m	200
Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM			Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM		
0.2	ohm.m	200	0.2	ohm.m	200
Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM			Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM		
0.2	ohm.m	200	0.2	ohm.m	200
Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC9 RM			Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC9 RM		
0.2	ohm.m	200	0.2	ohm.m	200

Environmentally Corrected (A40L) ARC9 RM		
0.2	ohm.m	200
Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM		
0.2	ohm.m	200
Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM		
0.2	ohm.m	200
Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM		
0.2	ohm.m	200
Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM		
0.2	ohm.m	200
Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC9 RM		
0.2	ohm.m	200

└─TICK_ARC_GR - Gamma Ray Tick Marks ARC9 RM

└─TICK_ARC_RES - Resistivity Tick Marks ARC9 RM

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:200 Index Unit: m Index Type: Measured Depth Creation Date: 12-Dec-2019 15:12:05

Channel Processing Parameters				
Run 3: Parameters				
Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC9	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	0	%
BS	Bit Size	DNMSESSION	Depth Zoned	in
DFD	Drilling Fluid Density	Borehole	1.03	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC9	2	ohm.m
LOW_BLEND	Low Resistivity Threshold for Blending	ARC9	1	ohm.m
RMS	Resistivity of Mud Sample	Borehole	0.19	ohm.m
Depth Zone Parameters				
Parameter	Value	Start (m)	Stop (m)	
BS	36	341	411.5	
BS	26	411.5	931	
All depth are actual.				
Tool Control Parameters				
Calibration Report				

ARC9 (Array Resistivity Compensated 900) Calibration - Run 3

Primary Equipment :

High Pressure without AIM Receiver

AREA

8374

RESAIRCAL - Resistivity: Air

Master (Time Frame File): 09:51:54 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	8.353	10.500	
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.149	8.500	
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	5.129	6.500	
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.142	6.600	
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.746	5.600	
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	0.713	4.100	
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-0.663	4.100	
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	0.666	4.100	
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-0.695	4.100	
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	0.651	4.100	
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	8.523	10.500	
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	5.988	8.500	
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	5.294	6.500	
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	3.975	6.600	
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.915	5.600	
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	0.360	4.100	
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	-0.415	4.100	
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	0.390	4.100	
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	-0.422	4.100	
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	0.384	4.100	

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File): 10:35:50 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.000	0.640	0.979	1.200	

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway



Schlumberger

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Dual Frequency
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